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The equity premium puzzle: empirical evidence for the “Korea Discount”

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This study examines the cost of equity capital (COC) and the factors that influence the COC of listed Korean companies, and compares the COC of Korean companies with that of companies from 31 selected countries. The major research question is whether companies that are listed in an emerging market have a disadvantage as they are underpriced through a higher COC (as compared with companies that are traded in developed markets). Consistent with the “Korea Discount,” we find that the COC is significantly higher for Korean companies than for companies in other countries after controlling for other relevant factors. However, the “Korea Discount” has significantly eased in recent years.

Keywords: analysts' forecasts; cost of capital; firm characteristics; Korea Discount; Korean stock market

JEL Codes: G00, G18, M20

Introduction

The purpose of this study is to examine the expected rates of return of companies that are listed in the Korean stock market by investigating: (1) the implied cost of capital (COC); (2) equity premia; and (3) factors that affect the expected rate of return.¹

A plethora of anecdotal evidence has shown the existence of the so-called “Korea Discount.” This discount refers to the valuation phenomenon whereby, in comparison with listed companies in other countries, listed Korean companies are, *ceteris paribus*, traded at a discount. Figure 1 illustrates that the price-to-earnings (P/E) and price-to-book (P/B) ratios for Korean companies are significantly lower than those for the rest of the sample, suggesting that listed Korean companies are substantially undervalued even when compared with companies from other Asian developing countries (Salmon 2007, Suh and Sim 2007). We observe the “Korea Discount” even though Korean companies exhibit greater profitability than companies in other countries. Concerning the “Korea Discount,” Guerrero (2006) commented:

HSBC analysts estimate that over the past 12 years, the Korean stock market had traded, on average, 26 per cent below the valuation of other exchanges in Asia excluding Japan.

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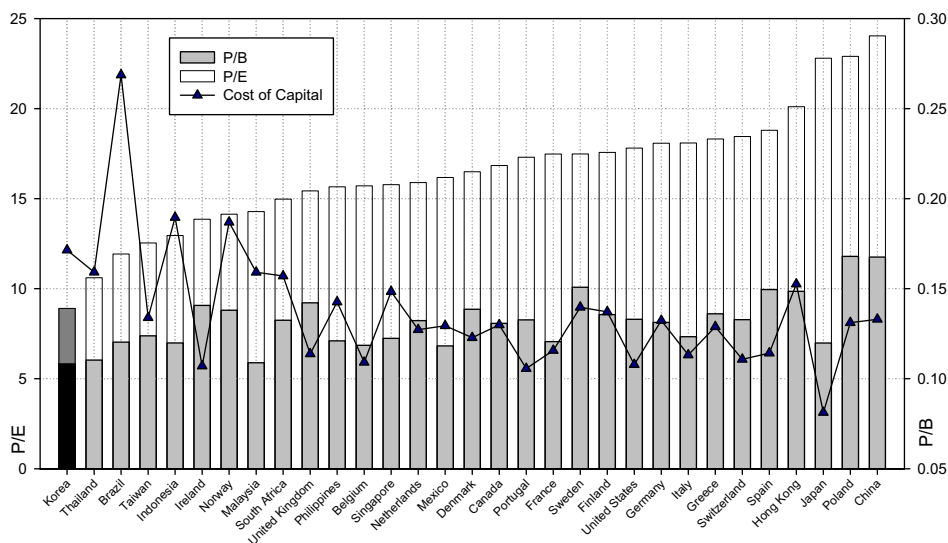


Figure 1. Country P/E ratio and P/B ratio.

In the late 1990s, Korea experienced serious economic difficulties. The collapse of the Thai baht in 1997 triggered a regional economic crisis that spread to other Asian countries, including Korea. Korean stock prices consequently plummeted, devastating the Korean market for several years. Since the economic turmoil, the attention Korea has been paying to corporate transparency has significantly increased and the Korean government has expressed great concern about the “Korea Discount.” Korean authorities have consequently taken a series of steps to restructure the Korean financial system (e.g. improving corporate governance and transparency) and, in early 2009, the Korean government even launched the national brand committee to mitigate the widespread “Korea Discount.”

In an effort to tackle the “Korea Discount” and increase market transparency, the Korean government decided to adopt the International Financial Reporting Standards (IFRS): all listed companies with assets greater than \$2 billion are required to adopt IFRS starting from 2011. Several companies early-adopted IFRS in 2009 (e.g. STX and Pulmuone holdings). Market participants, including analysts, investors, and financial authorities, expect that the adoption of IFRS will mitigate the market stigma of the “Korea Discount” by reducing uncertainty and the incidence of opaque accounting practices (Ramstad 2007a, Moon 2008).

The attention of Korean authorities has also been recently directed to promoting South Korea from the category of “emerging” markets to “developed status” in terms of the FTSE stock market index (the upgrade transpired in September 2009).² The government and market participants have expressed the hope that the status upgrade can abate the “Korea Discount” through the increased trust of investors in the Korean financial market. The status change also highlights the importance of the timing of this study as our findings can be helpful in determining the changes that accompany the status upgrade and early adoption of IFRS.

An investigation of the phenomenon of the “Korea Discount” would ponder several fundamental questions. Is the Korean stock market really undervalued after taking into

account differences in growth prospects? How do we measure the valuation difference? In our study, we seek to provide answers to the above questions in two steps. We first estimate the implied cost of capital for companies in the Korean stock market; then, we compare it with the COC for companies in other countries. Despite the contribution of previous literature, the link between the equity cost of capital and its effects on stock market undervaluation, such as the “Korea Discount,” is far from obvious. Having a direct effect on the present value of future economic benefits, the COC is a crucial factor in company valuation. Assuming the same long-term growth rate for all companies, on average, higher (respectively lower) firm value stems from a lower (respectively higher) COC. Hence, the valuation level of the stock market may be examined by estimating the implied costs of capital that are implicitly required by market participants after controlling for the expected growth prospects.

Another major research question is whether, when compared with companies in developed markets, companies that are listed in an emerging market (e.g. the Korean stock market) have a disadvantage as they are underpriced through higher implied costs of capital. We compare the COC of Korean companies with that of US companies and examine the exact nature of the relationship between the COC and the undervaluation of Korean corporations. The estimation of the COC in Korea and its comparison with the COC in the USA helps us to ascertain whether the COC in Korea can be regarded as a factor of the “Korea Discount.” In addition, we compare the risk premia of the two countries, since the equity risk premium is directly related to market valuation (Foerster and Karolyi 1999). Prior literature suggests that when they successfully reduce the aforementioned shortcomings of their financial system, emerging markets experience an upward revaluation of their shares (Miller 1999, Henry 2000), which is achieved mainly through the decrease of the COC (Henry 2000). We also investigate whether the undervaluation of the Korean stock market has eased up in recent years.

Finally, we estimate the COC for 32 countries and investigate its association with valuation measures after controlling for macro-economic variables such as the Gross Domestic Product (GDP), inflation, level of corruption and governance, and financial reporting quality.

Our results suggest that although a substantial country risk premium still exists in the Korean stock market, the implied costs of capital have decreased over recent years. These findings are in line with the argument that the “Korea Discount” has eased. Furthermore, we find a significant association between the measure of the implied cost of capital and variables that affect the risk and profitability, as perceived by investors. The implication of our findings may be extended to other countries.

Literature review

Extant literature dealing with the “Korea Discount” and its various aspects has pointed out several issues pertaining to the study of this phenomenon. One is the difficulty of measuring the extent of the “Korea Discount.” Numerous business articles have attempted to measure the level of the “Korea Discount” and speculated about its causes, but most of these studies have examined the levels of financial ratios, such as the P/E or P/B benchmarks. However, the validity of such studies depends on appropriate measures of valuation. Even though surveys of managers affirm the importance of the P/E and/or P/B ratios in valuing stocks (Shahed *et al.* 2008), the nature of the connection and the factors linking these ratios to stock-market valuation is less obvious (Kumar and Hyodo 2001, Bhargava and Malhotra 2006). The use of financial ratios in measur-

ing the level of the stock-market discount may lead to spurious inferences since their level may reflect not only the amount of risk but also other underlying economic factors of the concerned country (e.g. growth prospects, business sectors, interest rates, and size, etc.; Byrne and Saibeni 2000, Park 2000, Kumar and Hyodo 2001). Thus, for example, lower P/E or P/B ratios may indicate lower implied future growth prospects of companies in the country.³

The second issue is the choice of methodology in measuring the cost of capital. In many cases, academics and practitioners rely on the COC as estimated by methods based on historical data, such as the capital asset pricing model (CAPM). However, CAPM estimates have been recently criticized as having limited practical use (Fama and French 1997, 2002). To mitigate this problem, we estimate the expected rates of return by reverse-engineering the residual earnings model, which has been used by a large number of studies to test hypotheses related to the link between the required rates of return and relevant factors that may affect them (Dhaliwal *et al.* 2005, Daske 2006, Easton and Sommers 2007). In this study, we use Easton *et al.*'s (2002; henceforth ETSS) model to simultaneously estimate the growth rate and the COC; we also employ the Ohlson and Juettner-Nauroth (2005) model to estimate the firm-specific COC for 32 countries.

Only in the past decade have reverse-engineered valuation models been used to determine the COC; nevertheless, the number of studies that use the COC that is estimated from the reverse-engineered residual earnings or abnormal earnings growth model has recently been increasing (O'Hanlon and Steele 2000, Claus and Thomas 2001, Gebhardt *et al.* 2001, Easton *et al.* 2002, Baginski and Wahlen 2003, Gode and Mohanram 2003, Easton 2004, Easton and Monahan 2005, Easton and Sommers 2007). Having been developed to estimate the required rates of return at various levels of aggregation, reverse-engineered valuation methods can be used to make firm-specific estimates (Claus and Thomas 2001; Gebhardt *et al.* 2001, Baginski and Wahlen 2003, Gode and Mohanram 2003, Ohlson and Juettner-Nauroth 2005) as well as estimates for a group or portfolio (O'Hanlon and Steele 2000, Claus and Thomas 2001, Easton *et al.* 2002, Easton 2004). However, firm-specific estimation requires assumptions about the long-term growth rate (e.g. fading ROE to the industry median ROE (Gebhardt *et al.* 2001) or the same rate for the long-term growth of abnormal earnings for all companies (Claus and Thomas 2001, Gode and Mohanram 2003⁴), which can be regarded as a disadvantage of this approach.

The literature has also documented international differences in the COC and investigated the link between the required rate of return and potential factors (Agmon and Findlay 1982, Damodaran 2003, Chen *et al.* 2004, Koedijk and van Dijk 2004, Sabal 2004, Dargenidou *et al.* 2006, Daske 2006, Hail and Leuz 2006, He and Kryzanowski 2007). Several explanations have also been proposed regarding the questions of whether and why companies that are listed on the Korean stock market are considered to be undervalued,⁵ and a close link has been reported between undervaluation and corruption or political risk (i.e. North Korea) (Lee and Ng 2006, Pyo and Jung 2007, Pantzalis *et al.* 2008).

Research design

Firm-specific estimation of the implied cost of capital: the OJ model

In line with the main research question, our paper focuses on the estimation and comparison of the expected rates of return in Korea and other countries. COC and related issues have already been examined by numerous studies (Agmon and Findlay 1982,

Damodaran 2003, Chen *et al.* 2004, Koedijk and van Dijk 2004, Sabal 2004, Dargenidou *et al.* 2006, Daske 2006, Hail and Leuz 2006, He and Kryzanowski 2007, Choi and Jung 2008); however, a few determine the COC through reverse-engineered valuation models.

In our study, we choose to estimate the COC by the reverse-engineered valuation model because it enables us to estimate firm-specific COCs. We use these COC estimates in a cross-country comparison.⁶ Among the existing reverse-engineered models, we use the Ohlson and Juettner-Nauroth (2005) model as expressed below⁷:

$$r = A + \sqrt{A^2 + \frac{\text{eps}_1}{P_0}(g_s - (\gamma - 1))} \quad (1)$$

where $A \equiv \frac{1}{2}((\gamma - 1) + \text{dps}_1/P_0)$ and $g_s = \text{eps}_2 - \text{eps}_1/\text{eps}_1$ $\gamma = r_f - g_1 + 1$; γ =long-term earnings growth rate+1; g_s =short-term earnings growth rate; g_1 =long-term economic growth rate; r_f =yield on a three-year treasury bond; and eps_t =analysts' forecast of earnings per share at time t .

As previously mentioned, being relatively new, reverse-engineered valuation models have been increasingly used to determine the COC only since the second-half of the 1990s. The reverse-engineered residual earnings and abnormal earnings growth models have been used to estimate the expected rate of return on equity investment (O'Hanlon and Steele 2000, Claus and Thomas 2001, Gebhardt *et al.* 2001, Easton *et al.* 2002, Baginski and Wahlen 2003, Gode and Mohanram 2003, Easton 2004, Easton and Monahan 2005, Ohlson and Juettner-Nauroth 2005, Easton and Sommers 2007); and the COC determined by reverse-engineered models has also been employed to investigate the relationship between the required rates of return and the factors that potentially influence them (Dhaliwal *et al.* 2005, Cheng *et al.* 2006, Daske 2006, Easton and Sommers 2007). Previous literature has found that the COCs determined by different residual models are highly correlated and that the use of different residual models does not lead to significant differences in the results.⁸

Model estimation

Anecdotal evidence shows that the Korean stock market is undervalued; consequently, firms that are listed in the Korean stock exchange have to bear a significantly higher COC than their economic fundamentals warrant. In this section, we examine whether, after we control for macro-economic factors, the Korea Discount is due to a high COC.

Our empirical model includes those macro-economic variables that prior studies have found relevant to country valuation: inflation,⁹ GDP, Corruption Perception Index (CPI), country governance, and financial reporting quality. We expect the inflation variable to be positively correlated with the COC and negatively correlated with firm value. The GDP variable is represented by the natural logarithm of the GDP of the country expressed in millions of USD (Easterly 1993, Lee and Ng 2006).¹⁰ From a theoretical standpoint, GDP has a positive (negative) association with firm value (cost of capital). Extant literature has reported that companies listed in the stock markets of corrupted countries are undervalued (Lee and Ng 2006, Pantzalis *et al.* 2008), suggesting that the corruption level is a critical factor for country valuation. As a proxy for the level of corruption, we use the CPI provided by Transparency International.¹¹ The CPI index is a comprehensive measure of corruption based on data collected from up to 13 sources

including the IMD, World Bank, Political and Economic Risk Consultancy, and World Economic Forum, ranging from 1 (extremely corrupted) to 10 (extremely clean).

Prior studies report that firms from countries with stronger regulations and stricter law enforcement systems have lower COCs (Hail and Leuz 2006). We utilize the Worldwide Governance Indicators (WGI) issued by the World Bank, which covers 212 countries (Kaoufmann *et al.* 2008). The WGI comprises six different categories of measures that reflect the extent to which a country's governance system is functioning well: (1) voice and accountability (VA); (2) political stability and absence of violence (PV); (3) government effectiveness (GE); (4) regulatory quality (RQ); (5) rule of law (RL); and (6) control of corruption (CC).

According to the World Bank's webpage, the first dimension, VA, measures the "perceptions of the extent to which a country's citizens are able to participate in selecting their government, as well as freedom of expression, freedom of association, and a free media." The second dimension, PV, measures the "perceptions of the likelihood that the government will be destabilized or overthrown by unconstitutional or violent means, including politically-motivated violence and terrorism," and the third dimension, GE, deals with the "perceptions of the quality of public services, the quality of the civil service and the degree of its independence from political pressures, the quality of policy formulation and implementation, and the credibility of the government's commitment to such policies." RQ is defined as referring to the degree of "the ability of the government to formulate and implement sound policies and regulations that permit and promote private sector development," and RL as "the extent to which agents have confidence in and abide by the rules of society, and in particular the quality of contract enforcement, property rights, the police, and the courts, as well as the likelihood of crime and violence." Finally, the last dimension, CC, reflects the "extent to which public power is exercised for private gain, including both petty and grand forms of corruption, as well as 'capture' of the state by elites and private interests."¹²

Many studies have shown that financial reporting quality varies across countries and regions. Thus, we can assume that these differences may potentially influence the COC estimates through earnings forecasts and accounting measures and that lower reporting quality is likely to be accompanied by a significantly higher COC. To control for the effect of differences in financial reporting quality, our analysis includes control variables that represent proxies for financial quality. In line with the literature, we use three different categories of measures, viz. faithful representation of operating performance (FROP), Accounting Conservatism (AC), and quality of accruals (QA).

FROP, as measured by the discretionary accrual model below, reflects the extent of earnings management (i.e. management discretion over accruals). We assume that companies that display lower financial reporting quality will tend to engage in earnings management more extensively. In our study, the measures of FROP are the negatives of the median of absolute value of the estimated discretionary accruals.

$$\frac{TA_{it}}{A_{it-1}} = \beta_0 \frac{1}{A_{it-1}} + \beta_1 \frac{\Delta Rev_{it}}{A_{it-1}} + \beta_2 \frac{PPE_{it}}{A_{it-1}} + \beta_3 ROA_{it} + \varepsilon_{it} \quad (2)$$

where ΔRev_{it} = change in revenues; PE_{it} = property; A_{it-1} = total assets at the end of the preceding year; and ROA_{it} = return on assets at year t .

The discretionary or unexpected accruals (DA) are measured by the residuals of the accrual models.

$$DA_{it} \equiv \frac{TA_{it}}{A_{it-1}} - \hat{\beta}_0 \frac{1}{A_{it-1}} + \hat{\beta}_1 \frac{\Delta Rev_{it}}{A_{it-1}} + \hat{\beta}_2 \frac{PPE_{it}}{A_{it-1}} + \hat{\beta}_3 ROA_{it} \quad (3)$$

Regarding AC, the literature reports several different measures. Following Khan and Watts (2008), we employ a variation of Basu's measure, which requires us to divide our work into two stages. First, a cross-sectional regression is estimated for each year as shown below:

$$\begin{aligned} \frac{E_{it}}{P_{it-1}} = & \alpha_0 + \alpha_1 \cdot DR_{it} + \beta_{1t} \cdot SizeMVE_{it} + \beta_{2t} \cdot MTB_{it} + \beta_{3t} \cdot LEV_{it} + DR_{it} \cdot (\delta_0 \\ & + \delta_{1t} \cdot SizeMVE_{it} + \delta_{2t} \cdot MTB_{it} + \delta_{3t} \cdot LEV_{it}) + R_{it} \cdot (\gamma_0 + \gamma_{1t} \cdot SizeMVE_{it} \\ & + \gamma_{2t} \cdot MTB_{it} + \gamma_{3t} \cdot LEV_{it}) + R_{it} \times DR_{it} \cdot (\mu_0 + \mu_{1t} \cdot SizeMVE_{it} + \mu_{2t} \\ & \cdot MTB_{it} + \mu_{3t} \cdot LEV_{it}) + \varepsilon_{it} \end{aligned} \quad (4)$$

where E =net income scaled by the lagged market value of equity; R =annual stock returns for the 12-month period ending three months after the fiscal-period end; $SizeMVE$ =natural logarithm of the market value of equity; MTB =market-to-book ratio; and LEV =leverage measured by the sum of long-term and short-term debts deflated by the market value of equity.

In the second stage, we use the coefficient estimates from the first-stage regression, viz. (4), to calculate a firm-year specific measure of the timeliness of earnings with respect to conservatism (the C Score) at year-end. The measures of accounting conservatism that we seek are then represented by the medians of C Score.

$$C \text{ Score} \equiv \mu_{0t}^\wedge + \mu_{1t}^\wedge SizeMVE_{it} + \mu_{2t}^\wedge + MTB_{it} + \mu_{3t}^\wedge LEV_{it} \quad (5)$$

The third financial quality measure is the QA. In line with Burgstahler *et al.* (2006), we measure the quality of accruals of a firm by the negative of the median ratio of the absolute value of total accruals to the corresponding value of operating cash flows. This measure is similar to FROP in the sense that we assume that firms use accruals to manipulate operating cash flows. QA focuses on total accruals rather than the estimated discretionary accruals. This approach is simple and does not require an accrual model to estimate the non-discretionary accruals.

We also use two regression equations to test the hypothesis that, after the financial performance of the country (i.e. ROA) is controlled for, companies from countries with a weaker country governance structure, lower financial reporting quality, and higher level of corruption have a higher COC and a lower P/E ratio. After the macro variables (C_i) are controlled for, a country dummy variable (d_{Korea}) is used to capture the "Korea Discount."¹³ Finally, we also measure the differential impact of financial performance on the COC and the valuation by employing the interaction term of $d_{Korea} \times ROA$.

$$r_{jt} = \alpha_1 + \alpha_2 d_{Korea} + \beta_1 ROA_{jt} + \beta_2 d_{Korea} ROA_{jt} + \sum_{i=1}^n \gamma_i C_{i,jt} + \varepsilon_{jt} \quad (6)$$

$$\frac{P}{E}_{jt} = \alpha_1 + \alpha_2 d_{Korea} + \beta_1 ROA_{jt} + \beta_2 d_{Korea} ROA_{jt} + \beta_3 r_{jt} + \sum_{i=1}^n \gamma_i C_{i,jt} + \varepsilon_{jt} \quad (7)$$

where r_j =implied cost of equity capital for country j ; P/E_j =price-to-earnings ratio for country j ; d_{Korea} =country dummy (1=Korea, 0=other countries); ROA_j =return on

total assets for country j ; $C_{1,j}$ =annual inflation rate for country j ; $C_{2,j}$ =natural log of GDP in millions of US\$ for country j ; $C_{3,j}$ =corruption perception index (CPI) for country j ; $C_{4,j}$ =voice and accountability indicator for country j ; $C_{5,j}$ =political stability and absence of violence indicator for country j ; $C_{6,j}$ =government effectiveness indicator for country j ; $C_{7,j}$ =regulatory quality indicator for country j ; $C_{8,j}$ =rule of law indicator for country j ; $C_{9,j}$ =control of corruption indicator for country j ; $C_{10,j}$ =FROP for country j ; $C_{11,j}$ =accounting conservatism for country j ; and $C_{12,j}$ =QA for country j .

Data and sample selection

The financial data used in the study were collected from companies listed in the Korean stock market. We collected annual financial data for the period 2000 through 2007.¹⁴ Companies with a non-December fiscal year-end were excluded from the sample. The Korean sample consisted of companies that traded on either the Korea Stock exchange or the Korea Securities Dealers Automated Quotation (KOSDAQ) exchange.¹⁵

Measures for the financial variables were taken from financial statements and the stock market at the end of the fiscal year. All per share variables for Korean companies were adjusted for stock splits and stock dividends. Accounting data, including earnings per share, book value, sales, long-term debt, total assets, and number of shares were culled from the DataGuide Pro annual research files.¹⁶ Analysts' forecasts, stock prices, market betas, and risk-free rates (T -bond rates) were also obtained from DataGuide Pro database.¹⁷ Earnings forecasts were derived from the last consensus forecasts available in December.

The sample selection rule requires earnings forecasts for two years. For example, to be included in the sample of 2007, the last consensus forecasts of the sample firms available in December 2007 should include fiscal years ending on 31 December 2007 and 2008. The top and bottom percentiles of observations based on the COC, P/E, and P/B were simultaneously eliminated to mitigate the effect of extreme values. For the international comparison, accounting numbers and analysts' forecasts are culled from the I/B/E/S database, and risk-free rates (T -bond rates), GDP, and inflation rates are obtained from the Datastream database. The financial reporting quality is computed from the North American and global Compustat database. Finally, to investigate the impact of the country's corruption on the firm value and the COC, we use the CPI for the years 2000 through 2007 provided by Transparency International.¹⁸ The WGI index is from the World Bank (Kaoufmann *et al.* 2008). To be included in the international sample, each country is required to have at least 20 observations. Covering the years 2000 through 2007, the final sample consists of 8214 observations across 32 countries.

Empirical results

A variety of studies demonstrate that firm-specific risks affect the COC (Gebhardt *et al.* 2001, Gode and Mohanram 2003, Botosan and Plumlee 2005). Descriptive statistics of the variables used to measure the COC and firm characteristics are presented in Table 1. As the table shows, the median values of P/B (1.154) and P/E (10.375) are slightly lower than the corresponding historical averages in the US stock market. The median beta (0.815) is slightly lower than the market beta and the median values for ROA and ROE are 6.4 and 11.7%, respectively. The median value of Tobin's Q ratio is 1.065.

Temporal changes of the variables are summarized in Table 2. On average, the valuation variables of P/B, P/E, and Tobin's Q have increased over time. The implied costs

Table 1. Descriptive statistics of key variables.

Descriptive statistics	<i>N</i>	Min	Median	Max	Std.
P/B	2089	0.006	1.154	10.831	1.199
P/E	2034	0.238	10.375	909.091	36.537
Beta	1313	-0.349	0.815	2.351	0.332
D/M	2033	0.000	0.132	13.203	0.884
<i>D/A</i>	2033	0.000	0.086	0.740	0.107
ROA	2089	-0.190	0.064	0.414	0.057
ROE	2089	-1.146	0.117	0.613	0.090
Tobin's <i>Q</i>	2089	0.216	1.065	8.500	0.734
TA	2089	10.696	273.859	218,866.040	12,824.040

Notes: P/B: price-to-book value of equity; P/E: price-to-earnings ratio; TA: total assets in millions of Korean won; Beta: capital asset pricing model beta; D/M: debt-to-market capitalization; D/A: debt-to-total assets; ROA: return on total assets; ROE: return on common equity; Tobin's *Q* = $\frac{\text{Liability} + \text{Market value of equity}}{\text{Total assets}}$.

of capital in the early 2000s (i.e. in the immediate aftermath of the IMF bailout) were higher than those in more recent years. As would be expected, the implied costs of capital for Korean companies are higher than those for US companies. For Korean companies, *r* varies from a high of 24.5% in 2000 to a low of 17.6% in 2007. Ranging from a high of 16.5% in 2000 to a low of 12.3% in 2007, the estimated equity premium over the risk-free rate for the Korean stock market averages 15.1% over the years 2000 through 2007. This estimate of the equity premium for Korean companies is much higher than that for the US stock market reported in prior US studies: 2.5% over the years 1981 through 1995 (Gebhardt *et al.* 2001), 3.4% over the years 1985 through 1998 (Claus and Thomas 2001), or 5.3% over the years 1981 through 1998 (Easton *et al.* 2002).

The result may imply the existence of a significant discount in the Korean stock market. Although the implied cost of capital does not show a monotonic decrease over the years 2000 through 2007, it is apparent that the equity premium was much higher in the early 2000s (just after the Asian economic crisis). Such a finding seems reasonable because the required rates of investors were logically much higher during the economic turmoil as many companies had financial difficulties or went into default. A likely explanation for the decrease in recent years is that the "Korea Discount" has recently eased although the COC in Korea is still much higher than that in the US stock market. In sum, although there is an obvious difference in the estimation periods, similarly to Easton *et al.* (2002) we can observe that in comparison to Korea, the COC in the US stock market varies less across periods.

Table 3 presents the estimates of *r* for the Korean market based on the portfolios partitioned by industry group. Consistent with prior studies in the USA, the utility industry has a lower COC than in most other industries (Fama and French 1997, Gebhardt *et al.* 2001, Easton *et al.* 2002). That telecommunication services also show a low COC of 13.0%, which may reflect the monopolistic power of Korean telecommunication companies since just a few companies dominate the highly profitable market in Korea. The result is not surprising considering that prior literature has reported a negative association between market power and the COC (Cressy 1995). The highest cost of capital (23.7%) has been detected in the group of industrial goods.

Table 4 presents the Spearman correlation coefficients across the key variables. As would be expected, the table reveals significantly positive associations between

Table 2. Descriptive statistics by year: medians of key variables.

Descriptive statistics	2000	2001	2002	2003	2004	2005	2006	2007
r	0.245	0.230	0.219	0.214	0.211	0.208	0.187	0.176
T -bond	0.087	0.062	0.063	0.048	0.044	0.045	0.050	0.053
Equity premium	0.165	0.160	0.152	0.163	0.162	0.159	0.136	0.123
P/B	0.613	0.866	0.790	0.991	1.011	1.552	1.409	1.494
P/E	6.844	10.608	6.981	9.877	7.428	12.349	12.857	14.044
Beta	0.730	0.725	0.850	0.750	0.765	0.820	1.011	0.928
D/M	0.315	0.233	0.231	0.149	0.148	0.091	0.099	0.091
D/A	0.099	0.107	0.090	0.077	0.075	0.080	0.078	0.083
ROA	0.050	0.049	0.066	0.064	0.082	0.072	0.067	0.063
ROE	0.096	0.087	0.119	0.120	0.140	0.129	0.115	0.121
Tobin's Q	0.805	0.930	0.883	0.965	0.997	1.278	1.212	1.264
TA	376.042	175.262	295.181	296.568	310.359	271.483	230.040	310.714

Notes: r : implied cost of equity capital; equity premium: r - T -bond rate; P/B: price-to-book value of equity; P/E: price-to-earnings ratio; TA: total assets in millions of Korean won; Beta: capital asset pricing model beta; D/M: debt-to-market capitalization; D/A: debt-to-total assets; ROA: return on total assets; ROE: return on common equity; Tobin's $Q = \frac{\text{Liability} + \text{Market value of equity}}{\text{Total assets}}$.

Table 3. Annual estimates of the implied cost of equity capital (r) by industry.

Industry	R (%)	P/B	P/E	Beta	D/M	D/A	ROA	ROE	Tobin's Q
Telecommunication services	13.0	1.319	12.603	0.813	0.289	0.249	0.069	0.144	1.261
Consumer staples	14.6	1.293	12.971	0.665	0.119	0.087	0.065	0.118	1.144
utility	16.4	0.494	6.657	0.351	0.597	0.123	0.039	0.080	0.768
Consumer discretionary	19.9	1.251	10.576	0.791	0.125	0.072	0.071	0.128	1.120
Basic materials	21.4	0.672	9.406	0.807	0.343	0.124	0.048	0.083	0.835
Health care	21.8	1.319	11.678	0.615	0.118	0.077	0.076	0.112	1.146
Information technology	22.3	1.548	12.033	1.040	0.048	0.047	0.085	0.136	1.316
Energy	22.7	1.658	13.672	0.872	0.117	0.115	0.092	0.146	1.371
Industrials	23.7	1.038	10.500	0.933	0.254	0.141	0.052	0.101	1.007

Notes: P/B: price-to-book value of equity; P/E: price-to-earnings ratio; TA: total assets in millions of Korean won; Beta: capital asset pricing model beta; D/M: debt-to-market capitalization; D/A: debt-to-total assets; ROA: return on total assets; ROE: return on common equity; Tobin's $Q = \frac{\text{Liability} + \text{Market value of equity}}{\text{Total assets}}$.

valuation variables, and a negative association between valuation variables and risk measures except for the market beta. The estimated cost of capital is negatively correlated with P/B (-0.203), P/E (-0.075), and Tobin's Q (-0.185).

Table 5 presents the estimates of r for the portfolios partitioned by financial factors that represent firm characteristics. In this section, we compare the COC estimates and various firm characteristics to see how they are associated. The relationship between firm characteristics and COC is backed up by anecdotal evidence and also supported by prior studies that demonstrate that firm-specific risks affect the implied costs of capital (Gebhardt *et al.* 2001, Gode and Mohanram 2003, Botosan and Plumlee 2005).

The numerator of the P/B and P/E ratios, viz. market price, is based on the expected future earnings that market participants pay for (Ohlson 1995). If participants expect higher future economic benefits relative to the book value (earnings), the P/B (P/E) will show a higher value as the market's expectations are incorporated in the numerator. Prior literature has reported that low P/B firms face higher systematic risk and earn higher *ex post* returns than high P/B firms (Fama and French 1992, Berk *et al.* 1999). The stocks of low P/B firms should thus earn a higher risk premium.¹⁹ In general, the levels of P/B and P/E are negatively associated with the COC. COC estimates for the lowest quintile (portfolio 1) of P/B and P/E are 25.0 and 24.5%; for the highest quintile (portfolio 5), they are 19.0 and 22.5%, respectively. Tobin's Q also captures the value of companies. COC for the highest quintile of Tobin's Q (17.7%) is significantly lower than that for the lowest quintile (23.6%).

Total assets (TA), which represent the size effect of a firm, proxy for risk.²⁰ Prior studies document that investors' required rate of return is lower for larger companies, because these companies usually have dominant market power (Cressy 1995) and/or lower default risk. The literature on disclosure also argues that for larger firms, the degree of information asymmetry that exists between investors and companies can be diminished since the size also proxies for the availability of information. Hence, a negative association exists between the level of disclosure and the COC (Botosan 1997, Poshakwale and Courtis 2005, Habib 2006). The existence of a significant negative association between firm size and the implied costs of capital is supported by our findings, which show that the COC decreases from 23.7% for portfolio 1 to 17.6% for port-

Table 4. Correlations between firm characteristics.

	<i>r</i>	P/B	P/E	TA	Beta	D/M	D/A	ROA	ROE
P/B	−0.203 (<.0001)								
P/E	−0.075 (0.002)	0.528 (<.0001)							
TA	−0.208 (<.0001)	−0.169 (<.0001)	−0.012 (0.623)						
Beta	−0.033 (0.291)	0.344 (<.0001)	0.308 (<.0001)	0.343 (<.0001)					
D/M	0.182 (<.0001)	−0.584 (<.0001)	−0.273 (<.0001)	0.484 (<.0001)	−0.042 (0.180)				
D/A	0.085 (0.001)	−0.188 (<.0001)	−0.038 (0.125)	0.466 (<.0001)	0.150 (<.0001)	0.876 (<.0001)			
ROA	−0.244 (<.0001)	0.498 (<.0001)	−0.202 (<.0001)	−0.358 (<.0001)	−0.062 (0.044)	−0.615 (<.0001)	−0.427 (<.0001)		
ROE	−0.211 (<.0001)	0.536 (<.0001)	−0.275 (<.0001)	−0.157 (<.0001)	0.061 (0.049)	−0.394 (<.0001)	−0.195 (<.0001)	0.859 (<.0001)	
Tobin's <i>Q</i>	−0.185 (<.0001)	0.965 (<.0001)	0.493 (<.0001)	−0.216 (<.0001)	0.317 (<.0001)	−0.619 (<.0001)	−0.231 (<.0001)	0.552 (<.0001)	0.551 (<.0001)

Notes: This table shows Spearman correlations between firm characteristics. P/B: price-to-book value of equity; P/E: price-to-earnings ratio; TA: total assets in millions of Korean won; Beta: capital asset pricing model beta; D/M: debt-to-market capitalization; D/A: debt-to-total assets; ROA: return on total assets; ROE: return on common equity; Tobin's $Q = \frac{\text{Liability} + \text{Market value of equity}}{\text{Total assets}}$.

Table 5. Annual estimates of the implied cost of equity capital (r) by various financial factors.

Financial factor	Quintile of firm characteristics (%)				
	1st	2nd	3rd	4th	5th
P/B	25.0	22.1	20.7	18.9	19.0
P/E	24.5	20.6	18.9	19.3	22.5
TA	23.7	23.4	21.8	17.7	17.6
Beta	21.1	18.9	18.7	20.1	19.0
D/A	18.4	19.9	20.3	22.9	22.3
ROA	27.1	23.4	19.6	18.7	18.2
ROE	26.7	22.1	19.3	17.5	20.4
Tobin's Q	23.6	23.8	20.7	19.6	17.7

Notes: The partitions are formed by sorting the data into quintiles on the basis of the financial factors. For example, portfolio 1 includes the bottom 20% of observations based on the size of each financial factor. P/B: price-to-book value of equity; P/E: price-to-earnings ratio; TA: total assets in millions of Korean won; Beta: capital asset pricing model beta; D/M: debt-to-market capitalization; D/A: debt-to-total assets; ROA: return on total assets; ROE: return on common equity; Tobin's $Q = \frac{\text{Liability} + \text{Market value of equity}}{\text{Total assets}}$.

folio 5. On the other hand, it is known that small firms are more optimistically biased (Bhushan 1989, Brown 1999, Richardson *et al.* 2000, Easton and Sommers 2007) and may thus show upwardly biased COCs (Easton and Sommers 2007). It is thus extremely difficult, if not impossible, to empirically disentangle all those effects due to the complexity of the issue.

It is also well documented that the COC is positively correlated with firm risk. The measures we use to assess firm risk in this study are the market beta (Beta), debt-to-market (D/M), and debt-to-assets (D/A). In addition, we also use the CAPM beta to capture the firm-specific risk that is related to market volatility (Sharpe 1964, Lintner 1965, Mossin 1966). Given that, a positive association between Beta and the COC would be expected (Gordon and Gordon 1997, Gode and Mohanram 2003, Harris *et al.* 2003). Contrary to these expectations, although the pattern is less apparent, Beta and the COC are associated negatively. The value of r varies between 21.1% (portfolio 1) and 19.0% (portfolio 5), but stands at 18.7% for portfolio 3. The result is not surprising in the sense that the COCs that are estimated from the historic beta (i.e. the CAPM) are of limited use in predicting firm-specific risk.²¹

D/M and D/A measure the risk that is associated with the financial leverage of the firm. As the amount of debt in a firm's capital structure increases, so too does the risk that the firm takes on. This provides an incentive for market participants to require higher rates of returns for firms with higher ratios. We expect a positive association between the COC and financial leverage. It turns out that r for the lowest quintile of D/A is 18.4%; for the highest quintile, r is 22.3%.

ROA and ROE proxy the profitability of companies. According to our results, they are negatively associated with the COC. While r for the lowest quintiles of ROA and ROE is 27.1 and 26.7%, respectively, for the highest quintiles, it is 18.2 and 20.4%, respectively.

Apart from portfolio analysis, we also performed regression analysis to examine the impact of financial factors on the COC in various settings. The results, which are provided in Table 6, are consistent overall with the portfolio analysis reported in Table 5.

Table 6. Regression of the cost of capital on financial factors.

Variable	Pred. sign	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Intercept	±	0.438*** (17.26)	0.456*** (18.07)	0.459*** (18.24)	0.406*** (16.78)	0.419*** (17.23)	0.433*** (17.78)
ROA	−	−0.586*** (−4.42)	−0.782*** (−6.33)	−0.608*** (−4.35)	−0.139** (−2.13)	−0.252*** (−4.09)	−0.149** (−2.25)
ROE	−	−0.023*** (−4.19)			−0.028*** (−5.16)		
P/B	−		0.000* (1.65)		0.000* (1.90)		
P/E	−			−0.033*** (−2.93)			−0.046*** (−4.32)
Tobin's Q	−			−0.029*** (−7.66)			−0.028*** (−7.36)
TA	−	−0.028*** (−7.54)	−0.031*** (−8.22)	0.249*** (4.56)	−0.027*** (−7.27)	−0.030*** (−7.98)	0.314*** (6.00)
D/A	+	0.255*** (4.68)	0.246*** (4.49)	0.015 (0.92)	0.323*** (6.22)	0.336*** (6.41)	0.019 (1.19)
Beta	+	0.023 (1.41)	−0.001 (−0.03)	0.121	0.026 (1.61)	−0.002 (−0.11)	0.109
Adj. R^2		0.129	0.116		0.115	0.095	

Notes:

Model 1: $r_{it} = \alpha_1 + \beta_1 \text{ROA}_{it} + \beta_2 \text{P/B}_{it} + \beta_3 \text{TA}_{it} + \beta_4 D/A_{it} + \beta_5 \text{Beta}_{it} \varepsilon_{it}$.Model 2: $r_{it} = \alpha_1 + \beta_1 \text{ROA}_{it} + \beta_2 \text{P/E}_{it} + \beta_3 \text{TA}_{it} + \beta_4 D/A_{it} + \beta_5 \text{Beta}_{it} \varepsilon_{it}$.Model 3: $r_{it} = \alpha_1 + \beta_1 \text{ROA}_{it} + \beta_2 \text{Tobin's } Q_{it} + \beta_3 \text{TA}_{it} + \beta_4 D/A_{it} + \beta_5 \text{Beta}_{it} \varepsilon_{it}$.Model 4: $r_{it} = \alpha_1 + \beta_1 \text{ROE}_{it} + \beta_2 \text{P/B}_{it} + \beta_3 \text{TA}_{it} + \beta_4 D/A_{it} + \beta_5 \text{Beta}_{it} \varepsilon_{it}$.Model 5: $r_{it} = \alpha_1 + \beta_1 \text{ROE}_{it} + \beta_2 \text{P/E}_{it} + \beta_3 \text{TA}_{it} + \beta_4 D/A_{it} + \beta_5 \text{Beta}_{it} \varepsilon_{it}$.Model 6: $r_{it} = \alpha_1 + \beta_1 \text{ROE}_{it} + \beta_2 \text{Tobin's } Q_{it} + \beta_3 \text{TA}_{it} + \beta_4 D/A_{it} + \beta_5 \text{Beta}_{it} \varepsilon_{it}$.

r : implied cost of equity capital; P/E: price-to-earnings ratio; P/B: price-to-book value of equity; TA: Log(total assets in millions of Korean won); Beta: capital asset pricing model beta; D/A: debt-to-total assets; ROA: return on total assets; ROE: return on common equity; Tobin's $Q = \frac{\text{Liability} + \text{Market value of equity}}{\text{Total assets}}$. t -statistics are in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

While the first three models employ ROA to proxy for companies' profitability, the next three models use ROE. Not surprisingly, both independent variables show a significantly negative association with the COC. P/B, P/E, and Tobin's Q are used as firm-valuation proxies. As expected, all of them are significantly negatively associated with the COC. The results further suggest that TA has a negative impact on the COC and D/A has a positive association with the COC. The association between Beta and the COC is not significant at the conventional level.

Table 7 presents the COC by country. As would be expected, when compared with other countries, the implied costs of equity capital for Korean companies are higher.²² Among the 32 countries investigated, South Korea is ranked the fourth highest. Figure 1 also shows that Korea is suffering from substantially lower P/E and P/B ratios and a higher COC.

Table 8 reveals the relationship between the COC and key valuation variables including the trailing P/E, forward P/E, and PEG ratio. The trailing P/E ratio is often used by investors for security valuation. While the focus of its denominator is the historic value, the price in the numerator incorporates future prospects. Hence, the replacement of the reported earnings with earnings forecasts in the denominator leads to a forward P/E ratio. However, the ratio still implicitly assumes the future earnings growth to be zero. Knowing this shortcoming, a number of articles in the business press base their stock recommendation on the PEG ratio, which can be obtained by dividing the forward P/E ratio by the forecast growth. Our test that aims to determine whether countries with high values exhibit lower COCs includes all three alternative ratios.

The correlations between the COC estimate and all valuation variables are significantly negative: -0.421 for the trailing P/E, -0.400 for the forward P/E, and -0.934 for the PEG ratio. In line with our speculation, the result implies that countries with

Table 7. Implied cost of equity capital by country.

Country	<i>N</i>	<i>r</i> (%)	Country	<i>N</i>	<i>r</i> (%)
Japan	136	8.1	Poland	21	13.1
Portugal	27	10.6	Germany	454	13.2
Ireland	60	10.7	China	204	13.3
United States	2289	10.8	Taiwan	98	13.4
Belgium	146	10.9	Finland	254	13.7
Switzerland	258	11.1	Sweden	362	14.0
Italy	249	11.3	Philippines	50	14.3
UK	804	11.4	Singapore	239	14.8
Spain	174	11.4	Hong Kong	62	15.3
France	473	11.6	South Africa	45	15.7
Denmark	125	12.3	Malaysia	256	15.9
Netherlands	258	12.7	Thailand	325	15.9
Greece	119	12.9	Korea	33	17.1
Mexico	25	12.9	Norway	180	18.7
Canada	311	13.0	Indonesia	74	19.0
Austria	52	13.0	Brazil	51	26.9

Notes: This table shows the implied cost of equity capital by country. The Ohlson and Juettner-Nauroth (2005) model (OJ model) is used to estimate the cost of equity capital. $r = A + \sqrt{A^2 + \text{eps}_1/P_0(g_s - (\gamma - 1))}$ where $A \equiv 1/2((\gamma - 1) + \text{dps}_1/P_0)$ and $g_s = \text{eps}_2 - \text{eps}_1/\text{eps}_1$; $\gamma = r_f - g_1 + 1$; γ = long-term earnings growth rate + 1; g_s = short-term earnings growth rate; g_1 = long-term economic growth rate; r_f = yield on a three-year treasury bond; and eps_t = analysts' forecast of earnings per share at time t .

Table 8. Country-level correlations between the implied cost of equity capital, valuation variables, and control variables.

	<i>r</i>	Trailing P/E	Forward P/E	PEG	Inflation	GDP	CPI	VA	PV	GE	RQ	RL	CC	FROP	AC
Trailing P/E	-0.421 (0.016)														
Forward P/E	-0.400 (0.023)	0.827 (<0.001)													
PEG	-0.934 (<0.001)	0.515 (0.003)	0.428 (0.015)												
Inflation	0.288 (0.137)	-0.366 (0.056)	-0.420 (0.026)	-0.244 (0.211)											
GDP	-0.206 (0.266)	0.150 (0.419)	0.088 (0.640)	0.146 (0.433)	-0.051 (0.797)										
CPI	-0.004 (0.982)	-0.041 (0.823)	0.056 (0.763)	-0.084 (0.647)	-0.582 (0.001)	0.119 (0.522)									
VA	-0.448 (0.010)	0.142 (0.437)	0.238 (0.189)	0.360 (0.043)	-0.574 (0.001)	-0.018 (0.923)	0.610 (0.000)								
PV	-0.209 (0.250)	0.155 (0.397)	0.339 (0.057)	0.098 (0.596)	-0.745 (<0.001)	0.090 (0.629)	0.669 (<0.001)	0.672 (<0.001)							
GE	-0.173 (0.343)	0.044 (0.810)	0.140 (0.443)	0.063 (0.732)	-0.742 (<0.001)	0.112 (0.550)	0.822 (<0.001)	0.719 (<0.001)	0.861 (<0.001)						
RQ	-0.351 (0.049)	0.082 (0.654)	0.198 (0.278)	0.215 (0.237)	-0.526 (0.004)	0.104 (0.578)	0.742 (<0.001)	0.688 (<0.001)	0.806 (<0.001)	0.887 (<0.001)					
RL	-0.239 (0.188)	0.077 (0.674)	0.207 (0.255)	0.116 (0.527)	-0.689 (<0.001)	0.135 (0.469)	0.788 (<0.001)	0.809 (<0.001)	0.882 (<0.001)	0.960 (<0.001)	0.872 (<0.001)				
CC	-0.238 (0.191)	0.308 (0.086)	0.317 (0.077)	0.126 (0.493)	-0.720 (<0.001)	-0.131 (0.483)	0.833 (<0.001)	0.809 (<0.001)	0.878 (<0.001)	0.963 (<0.001)	0.904 (<0.001)	0.976 (<0.001)			
FROP	-0.586 (0.000)	0.170 (0.352)	0.072 (0.694)	0.314 (0.080)	-0.021 (0.917)	-0.175 (0.346)	-0.219 (0.229)	0.174 (0.341)	0.041 (0.826)	-0.145 (0.428)	-0.078 (0.671)	-0.061 (0.740)	-0.047 (0.800)		
AC	-0.248 (0.172)	-0.020 (0.913)	0.005 (0.978)	0.316 (0.078)	-0.083 (0.673)	0.290 (0.113)	0.143 (0.435)	0.209 (0.251)	0.025 (0.894)	0.159 (0.386)	0.205 (0.261)	0.103 (0.573)	0.148 (0.419)	-0.169 (0.356)	
QA	-0.297 (0.099)						0.333 (0.063)	0.555 (0.001)	0.265 (0.143)	0.325 (0.070)	0.323 (0.072)	0.404 (0.022)	0.438 (0.012)	0.382 (0.031)	-0.062 (0.737)

Notes: This table presents Spearman correlations between key variables: Trailing P/E = Price/Reported earnings per share; Forward P/E = Price/Reported earnings per share_{t+1}; PEG = Trailing P/E/Short-term growth of earnings_{t+1}; *r*: implied cost of equity capital estimated by the OJ Model; Inflation: annual inflation rate of the country; GDP: natural log of the GDP of the country in millions of US\$; CPI: corruption perception index; VA: voice and accountability; PV: political stability and absence of violence; GE: government effectiveness; RQ: regulatory quality; RL: rule of law; CC: control of corruption; FROP: faithful representation of operating performance; AC: accounting conservatism; QA: quality of accruals.

Table 9. Regression analysis of the cost of capital and P/E ratios: Korea vs. other countries.

Variable	Predicted sign	Model 1	Model 2	Model 3	Predicted sign	Model 4	Model 5	Model 6
Intercept	±	0.131*** (103.63)	0.205*** (20.69)	0.234*** (22.63)	+	19.399*** (35.34)	10.538*** (4.39)	8.867*** (3.47)
d_{Korea}	+	0.083** (2.38)	0.077** (2.28)	0.061* (1.83)	−	−16.276** (−2.12)	−14.990* (−1.95)	−17.151** (−2.22)
ROA	−	0.027*** (2.54)	−0.002 (−0.24)	−0.007 (−0.69)	+	14.753 (6.22)	16.880*** (7.09)	17.175*** (7.21)
ROA _{Korea}	±	−0.562** (−2.06)	−0.559** (−2.16)	−0.555** (−2.16)	±	25.322 (0.42)	24.662 (0.42)	28.979 (0.49)
r					−	−11.460*** (−3.18)	−6.343* (−1.68)	−5.365 (−1.41)
Inflation	+		+0.282** (2.47)	0.298*** (2.62)	−	−3.328 (−0.13)	−3.328 (−0.13)	−6.924 (−0.26)
GDP	−		−0.001 (−0.71)	−0.001 (−0.73)	+	0.899*** (3.81)	0.899*** (3.81)	0.964*** (4.08)
CPI	−		−0.004*** (−5.23)	−0.003*** (−3.65)	+	−0.400** (−2.32)	−0.400** (−2.32)	−0.431** (−2.44)
VA	−		−0.016*** (−2.81)	−0.009*** (−2.81)	+	1.184* (1.78)	1.184* (1.78)	2.107*** (2.74)
PV	−		0.013** (2.56)	0.013** (2.57)	+	3.034*** (2.66)	3.034*** (2.66)	1.998* (1.71)
GE	−		0.015 (1.60)	0.010 (1.13)	+	−3.089 (−1.48)	−3.089 (−1.48)	−1.537 (−0.72)
RQ	−		−0.056*** (−7.49)	−0.062*** (−8.11)	+	−0.358 (−0.21)	−0.358 (−0.21)	−0.670 (−0.38)
RL	−		−0.030*** (−2.70)	−0.019* (−1.70)	+	−4.109 (−1.62)	−4.109 (−1.62)	−4.915* (−1.93)
CC	−		0.043*** (6.00)	0.041*** (5.65)	+	4.915*** (2.95)	4.915*** (2.95)	5.391*** (3.23)
FROP	−			−0.001*** (−7.05)	+			0.105*** (3.44)
AC	−			−0.001*** (−4.87)	+			−0.005 (−0.15)
QA	−			−0.0005* (−1.68)	+			−0.117*** (−3.22)
Adj. R^2		0.002	0.102	0.121		0.014	0.031	0.035

Notes:

Model 1: $r_{jt} = \alpha_1 + \alpha_2 d_{Korea} + \beta_1 ROA_{jt} + \beta_2 d_{Korea} + \varepsilon_{jt}$.Model 2: $r_{jt} = \alpha_1 + \alpha_2 d_{Korea} + \beta_1 ROA_{jt} + \beta_2 d_{Korea} + \beta_3 r_{jt} + \gamma_1 Inflation_{jt} + \gamma_2 GDP_{jt} + \gamma_3 CPI_{jt} + \gamma_4 VA_{jt} + \gamma_5 PV_{jt} + \gamma_6 GE_{jt} + \gamma_7 RQ_{jt} + \gamma_8 RL_{jt} + \gamma_9 CC_{jt} + \varepsilon_{jt}$.Model 3: $r_{jt} = \alpha_1 + \alpha_2 d_{Korea} + \beta_1 ROA_{jt} + \beta_2 d_{Korea} + \beta_3 r_{jt} + \gamma_1 Inflation_{jt} + \gamma_2 GDP_{jt} + \gamma_3 CPI_{jt} + \gamma_4 VA_{jt} + \gamma_5 PV_{jt} + \gamma_6 GE_{jt} + \gamma_7 RQ_{jt} + \gamma_8 RL_{jt} + \gamma_9 CC_{jt} + \gamma_{10} FROP_{jt} + \gamma_{11} AC_{jt} + \gamma_{12} QA_{jt} + \varepsilon_{jt}$.Model 4: $P/E_{jt} = \alpha_1 + \alpha_2 d_{Korea} + \beta_1 ROA_{jt} + \beta_2 d_{Korea} + \beta_3 r_{jt} + \varepsilon_{jt}$.Model 5: $P/E_{jt} = \alpha_1 + \alpha_2 d_{Korea} + \beta_1 ROA_{jt} + \beta_2 d_{Korea} + \beta_3 r_{jt} + \gamma_1 Inflation_{jt} + \gamma_2 GDP_{jt} + \gamma_3 CPI_{jt} + \gamma_4 VA_{jt} + \gamma_5 PV_{jt} + \gamma_6 GE_{jt} + \gamma_7 RQ_{jt} + \gamma_8 RL_{jt} + \gamma_9 CC_{jt} + \varepsilon_{jt}$.Model 6: $P/E_{jt} = \alpha_1 + \alpha_2 d_{Korea} + \beta_1 ROA_{jt} + \beta_2 d_{Korea} + \beta_3 r_{jt} + \gamma_1 Inflation_{jt} + \gamma_2 GDP_{jt} + \gamma_3 CPI_{jt} + \gamma_4 VA_{jt} + \gamma_5 PV_{jt} + \gamma_6 GE_{jt} + \gamma_7 RQ_{jt} + \gamma_8 RL_{jt} + \gamma_9 CC_{jt} + \gamma_{10} FROP_{jt} + \gamma_{11} AC_{jt} + \gamma_{12} QA_{jt} + \varepsilon_{jt}$.

r : implied cost of equity capital; P/E: price-to-earnings ratio; d_{Korea} : 1 = Korea, 0 = other countries; ROA: return on total assets; ROA_{Korea} = return on total assets for Korean companies; Inflation: annual inflation rate of the country; GDP: natural log of the GDP of the country in millions of US\$; CPI: corruption perception index; VA: voice and accountability; PV: political stability and absence of violence; GE: government effectiveness; RQ: regulatory quality; RL: rule of law; CC: control of corruption; FROP: faithful representation of operating performance; AC: accounting conservatism; QA: quality of accruals. t -statistics are in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

high COCs are significantly undervalued. Nationally, we examine whether higher corruption is associated with lower country value through a higher COC. Prior literature documents that corruption has an adverse impact on the values of companies (Lee and Ng 2006, Pantzalis *et al.* 2008). Consistent with prior studies, the CPI shows a significantly negative correlation with the COC and a positive correlation with the valuation variables although in this case the significance level is low except for the forward P/E ratio. The impact of inflation and GDP on valuation is less strong. Prior literature reports that firms from countries with better regulations and law enforcement systems have lower COCs (Hail and Leuz 2006). In general, the WGI are negatively correlated with the implied cost of capital. In particular, voice and accountability (-0.448) and regulatory quality (-0.351) show significantly negative correlations with the COC. Another important factor for the cross-country study is the financial reporting quality, which is known to vary across countries as well as regions. The COC is negatively correlated with FROP (-0.586), AC (-0.248), and QA (-0.297). However, although the correlation between the COC and AC is of the predicted sign, it is insignificant at the conventional level.

Finally, Table 9 presents the results of regression analysis where the COC is the dependent variable and ROA is the independent variable, after the macro variables are controlled for. Following prior studies, we used inflation, GDP, CPI, WGI, and financial reporting quality as the controlling variables.²³ The results indicate that after the financial performance and macro variables are controlled for, the level of COC for the Korean stock market (0.061) is significantly higher in comparison with those in other countries. Interestingly enough, the COC for the Korean stock market is significantly negatively associated with ROA, implying that the financial performance of Korean firms has a substantial impact on the COC. In Model 3, the macro variables represented by *Inflation* (0.298), GDP (-0.001), and CPI (-0.003) also follow the predicted signs. The association between r and WGI shows mixed results. While voice and accountability (-0.009), regulatory quality (-0.062), and rule of law (-0.019) follow the predicted sign, political stability and absence of violence (0.013) and control of corruption (0.041) show the opposite sign. Government effectiveness is insignificant. Model 6 shows the results of regression using the P/E ratio as the dependent variable, and ROA and the COC as the independent variables, after controlling for the macro variables.²⁴ The association between the COC and the company's valuation (P/E) is highly significant. Such a conclusion is not surprising given that the COC has an immediate impact on the stock price. As expected, after the financial performance, COC, and macro variables are controlled for, the P/E ratio of the Korean stock market (-17.151) is significantly lower than that of other countries. From the test, we can conclude that the Korean stock market is substantially undervalued in comparison with other stock markets.

Concluding remarks

A variety of evidence has shown the existence of the "Korea Discount," referring to the phenomenon that the companies listed in the Korean stock market are, *ceteris paribus*, traded at a discount in comparison with companies in other countries. The market stigma of the "Korea Discount" has spread widely to other economic areas including the national brand value and foreign direct investment (FDI).²⁵

The results of this research have several important implications for the study of equity premia. First, this paper provides compelling evidence that the Korean stock market is traded at a discount. The estimated equity premium over the risk-free rate for

the Korean stock market averages 15.1% over the years 2000 through 2007, which is significantly higher than that reported for the US stock market. Second, our study shows that the equity premium has decreased over the last eight years. We postulate that the economic turmoil of the 1997 financial crisis negatively (in a valuation sense) affected the Korean stock market in several subsequent years and investors' required rates were much higher after the crisis as many companies had financial difficulties if not already in default.

However, the "Korea Discount" has eased up in recent years. It appears that the market disadvantage has decreased since the late 1990s, partly due to the efforts of the government to enhance corporate transparency including early adoption of the IFRS or advancement to "developed status" according to the FTSE stock market index. It is also expected that the status promotion to "advanced" country based on the MSCI index, which is supposed to be decided in June 2010, will also provide a positive signal to potential investors regarding the level of development and safety of the Korean financial market.

As well documented in prior studies, we found a significant association between firm characteristics and the COC (Gebhardt *et al.* 2001, Gode and Mohanram 2003, Botosan and Plumlee 2005). To investigate this issue, we examined a measure of the implied cost of capital in relation to variables that affect the risk and profitability, as perceived by market investors.

Finally, after controlling for macro variables such as GDP, inflation, a country's level of corruption, the WGI, and financial reporting quality, we estimated the COC for 32 countries and found that the Korean stock market is substantially undervalued.

Nevertheless, the results of this study should be interpreted with some caveats. One issue to be wary of is the potential effect of omitted firm-level variables on the estimated COC. For example, firm-level disclosure and/or information quality may have a significant impact on the implied cost of capital (Diamond and Verrecchia 1991, Botosan and Plumlee 2002, Easley and O'Hara 2004, Berger *et al.* 2006, Core *et al.* 2007, Lambert *et al.* 2007).

Another issue is the potential upward bias in the estimates of the costs of capital implied by market prices and analysts' forecasts. The high costs of capital may reflect the fact that earnings forecasts are optimistically biased. If so, the COC that is implied by inflated analysts' forecasts of earnings will be upwardly biased as well (Easton and Sommers 2007). A related problem is that earnings forecasts that are prepared under different accounting regimes (e.g. GAAP or Korean GAAP) may lead to differences in the COC (Easton 2006); different GAAP may thus affect estimates of r .

Future research can fruitfully extend this study in several directions. One valuable area for future study would be an analysis of analysts' optimism and the COC using a dataset that spans a longer time-window. Besides, since caution must be exercised when we interpret omitted firm-level variables, further research is required to investigate the link between disclosure or information quality and the phenomenon of stock market discounts. Another worthwhile area of research would be one that looks into cross-sectional and longitudinal analyses of analysts' forecast errors for Korean companies and investigates the extent to which forecast errors are attributable to the COC. Further investigation of cross-cultural differences among countries would also lead to fruitful results.

Overall, this study is among the first to examine the implied equity premium for the Korean stock market. Our research on the links between the implied cost of capital and

firm characteristics sheds more light on this phenomenon and provides another potent interpretation of the “Korea Discount” based on the equity premium.

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Notes

1. Estimates of the cost of capital are implied by market prices, analysts' forecasts of earnings, and accounting numbers. Despite not being exactly identical, the terms, 'implied cost of capital' and 'expected rate of return,' are often used as equivalent to the cost of equity capital. They are the same only if we assume efficient market prices, and unbiased accounting numbers and analysts' forecasts.
2. Ramstad (2007b) noted: "... Even after South Korea met developed-market income qualifications in 2003, the firms that create indexes expressed concerns from market volatility to weak dividend payouts to the risk of trouble from neighboring North Korea ... An upgrade in market status has become a major goal for some government officials and news media in South Korea, where rankings and comparisons with other countries are taken very seriously. After FTSE and MSCI criticized the low level of dividends paid by South Korean companies, the government, which owns the Korean stock xchange, in 2004 encouraged companies to change by creating an index of the top 50 companies based on dividends."
3. Knowing the weakness of the P/E ratio, market practitioners frequently use other arguable heuristics such as the price-to-sales (P/S) or the PEG ratio (Bradshaw 2002, Fisher 2007). Nevertheless, even these ratios are still significantly affected by other economic factors (Easton 2004).
4. E.g. the expected inflation rate: risk-free rate -3% .
5. One line of reasoning is that the Korean financial system has shortcomings such as: (1) volatility of the Korean stock market due to investors' characteristics regarding short-term speculation (Chang 2005); (2) restrictions on the stock market (e.g. restricted short-selling); and (3) general weakness of financial systems (e.g. insufficient shareholder protection, restrictions on hedge or pension funds, etc.). Besides, various serious discount factors represented by: (1) poor corporate governance (Baek *et al.* 2004, Hail and Leuz 2006, Baek *et al.* 2009); (2) lack of business ethics, including corporate transparency (Baek *et al.* 2004, Choi and Jung 2008, Choi and Nakano 2008); (3) inadequate and less timely disclosure (Botosan 1997, Botosan and Plumlee 2002, Poshakwale and Courtis 2005, Dargenidou *et al.* 2006, Habib 2006, Lambert *et al.* 2007); (4) militant labor unions (Seggerman 2007); and (5) low national brand value have also been provided by business corporations. Concerning the undervaluation of Korea, Kim (2008) reports: "According to the 2007 Anholt-GMI Nation Brands Index, Korea ranked 32nd out of 38 countries worldwide in terms of national brand. Korea's national brand value accounts for merely 29% of its gross domestic product, which is paltry compared with 224% for Japan and 143% for the USA. What is worse, Korea lags behind countries with smaller economies, like Mexico, Poland and Egypt, in terms of national brand value. For that reason, Korean products are more than 30% cheaper than similar quality products from Germany and Japan in international markets."
6. The COC was also estimated by using different models including the modified PEG ratio model (O'Hanlon and Steele 2000, Gebhardt *et al.* 2001, Easton *et al.* 2002, Easton 2004). Implied by the prior discussion, regarding the validity of COC, the mean (median) value of COC estimated by alternative models was used as a proxy for the cost of capital (Hail and Leuz 2006) in addition to individual estimates. The correlation between the COCs that were estimated by alternative models was very high and the results were qualitatively similar.
7. We also estimated the COC from a PEG model (Easton 2004) as follows: $\sqrt{\text{eps}_2 - \text{eps}_1/P_0}$. The result was not sensitive to the choice of the estimation method.
8. For example, when Hail and Leuz (2006) estimated the COC by using four different models, they found that the choice of model did not have a significant impact on the results.
9. Annual inflation rate (Nissim and Penman 2003).

10. We also used the GDP growth rate as well as the GDP without the natural log operator. The results were qualitatively identical.
11. www.transparency.org.
12. www.worldbank.org.
13. We also used firm-level measures for the quality of accruals. The result was qualitatively similar.
14. The archival database for earnings forecasts for Korean companies was available from 2000.
15. As of 31 December 2007, 745 companies were listed on the Korean stock exchange and 1026 companies were listed on the KOSDAQ exchange.
16. DataGuide Pro, which is a financial database prepared and maintained by FnGuide, Inc., provides companies' financial data, analysts' reports, and economic data. TS2000 and KSRI were also used as alternative sources of financial data. The results were qualitatively similar. TS2000 is a database that provides companies' financial data; it is prepared and maintained by the Korea Listed Companies Association. The KSRI Stock Database is provided by the Korea Securities Research Institute.
17. We use financial data from DataGuide Pro as this database contains a complete dataset for Korean companies; I/B/E/S reports substantially fewer observations.
18. www.transparency.org.
19. Gebhardt *et al.* (2001) report that P/B is the most important variable in explaining the variation in the implied costs of capital.
20. Alternative financial measures including market capitalization were also tested as proxy variables.
21. Fama and French (1997, 2002) conclude that the cost of capital estimated from historic beta is unavoidably imprecise.
22. We report the COC as estimated by the OJ model since the data used by this model are more easily available; thus, we were able to obtain more observations. We speculate that the implications of an analysis based on the estimates obtained by the model of Easton *et al.* (2002) will be qualitatively similar as prior literature has reported a very high correlation between the costs of capital that are legitimated by these two methods (e.g. Daske 2006).
23. We performed the analysis with various control variables such as the market beta, total assets, and ROE. The implication of the result was not sensitive to the choice of control variables.
24. The analysis was also conducted with the P/B ratio. The result was qualitatively similar.
25. For example, by 2007, Korea had slipped for three consecutive years in terms of FDI and ranked 60th globally.

References

- Agmon, T. and Findlay, M.C., 1982. Domestic political risk and stock valuation. *Financial analysts journal*, 38, 74–77.
- Baek, H., Johnson, D., and Kim, J., 2009. Managerial ownership, corporate governance, and voluntary disclosure. *Journal of business & economic studies*, 15 (2), 44–61.
- Baek, J., Kang, J., and Park, K., 2004. Corporate governance and firm value, evidence from the Korean financial crisis. *Journal of financial economics*, 71, 265–313.
- Baginski, S. and Wahlen, J., 2003. Residual income, risk, intrinsic values, and share prices. *The accounting review*, 78, 327–351.
- Berger, P., Chen, H., and Li, F., 2006. *Firm specific information and cost of equity*. Available from SSRN: <http://ssrn.com/abstract=906152>.
- Berk, J., Green, R., and Naik, V., 1999. Optimal investment, growth options, and security returns. *Journal of finance*, 54, 1553–1607.
- Bhargava, V. and Malhotra, D., 2006. Do price-earnings ratios drive stock value?. *The journal of portfolio management*, (Fall), 86–114.
- Bhushan, R., 1989. Firm characteristics and analysts following. *Journal of accounting and economics*, 11, 256–274.
- Botosan, C., 1997. Disclosure level and the cost of equity capital. *The accounting review*, 72, 323–349.
- Botosan, C. and Plumlee, M., 2002. A re-examination of disclosure level and the expected cost of equity capital. *Journal of accounting research*, 40, 21–40.

- Botosan, C. and Plumlee, M., 2005. Assessing alternative proxies for the expected risk premium. *The accounting review*, 80, 21–54.
- Bradshaw, M., March 2002. The use of target prices to justify sell-side analysts' stock recommendation. *Accounting horizons*, 16 (1), 27–41.
- Brown, L., 1999. *Managerial behavior and the bias in analysts' earnings forecasts*. Working paper. Georgia State University.
- Burgstahler, D., Hail, L., and Leuz, C., 2006. The importance of reporting incentives: earnings management in European private and public firms. *The accounting review*, 81, 983–1016.
- Byrne, G. and Saibeni, A., 2000. Stock value: the price/earnings fallacy. *Business forum*, 25, 7–10.
- Chang, K., 2005. Volatility, risk premium and Korea discount. *The Korean journal of financial management*, 22, 165–187.
- Chen, F., Jorgensen, B.N., and Yoo, Y.K., 2004. Implied cost of equity capital in earnings-based valuation: international evidence. *Accounting and business research*, 34, 323–344.
- Cheng, C., Collins, D., and Huang, H., 2006. Shareholder rights, financial disclosure and the cost of equity capital. *Review of quantitative finance and accounting*, 27, 175–204.
- Choi, T. and Jung, J., 2008. Ethical commitment, financial performance, and valuation: an empirical investigation of Korean companies. *Journal of business ethics*, 81, 447–463.
- Choi, T. and Nakano, C., 2008. The evolution of business ethics in Japan and Korea over the last decade. *Human systems management*, 27 (3), 183–199.
- Claus, J. and Thomas, J., 2001. Equity premia as low as three percent?: empirical evidence from analysts' earnings forecasts for domestic and international stock markets. *Journal of finance*, 56, 1629–1666.
- Core, J., Guay, W., and Verdi, R., 2007. *Is accruals quality a priced risk factor?* Available from SSRN: <http://ssrn.com/abstract=911587>
- Cressy, R., 1995. Cost of capital and market power: the effect of size dispersion & entry barriers on market equilibrium. *Small business economics*, 7, 205–212.
- Damodaran, A., 2003. Country risk and company exposure: theory and practice. *Journal of applied finance*, 13, 63–76.
- Dargenidou, C., McLeay, S., and Raonic, I., 2006. Expected earnings growth and the cost of capital: an analysis of accounting regime change in the European finance market. *Abacus*, 42, 388–414.
- Daske, H., 2006. Economic benefits of adopting IFRS or US-GAAP – have the expected cost of equity capital really decreased? *Journal of business finance and accounting*, 33, 329–373.
- Dhaliwal, D., Krull, L., Li, O., and Moser, W., 2005. Dividend taxes and implied cost of equity capital. *Journal of accounting research*, 43, 675–708.
- Diamond, D. and Verrecchia, R., 1991. Disclosure, liquidity, and the cost of capital. *Journal of finance*, 46, 1325–1359.
- Easley, D. and O'Hara, M., 2004. Information and the cost of capital. *Journal of finance*, 59 (4), 1553–1583.
- Easterly, W., 1993. Good policy or good luck? Country growth performance and temporary shocks. *Journal of monetary economics*, 32, 459–483.
- Easton, P., 2004. PE ratios, PEG ratio, and estimating the implied expected rate of return on equity capital. *The accounting review*, 79, 73–96.
- Easton, P., 2006. Use of forecasts of earnings to estimate and compare cost of capital across regimes. *Journal of business, finance, and accounting*, 33, 374–394.
- Easton, P. and Monahan, S., 2005. An evaluation of accounting-based measures of expected returns. *The accounting review*, 80, 501–538.
- Easton, P. and Sommers, G., 2007. Effect of analysts' optimism on estimates of the expected rate of return implied by earnings forecasts. *Journal of accounting research*, 45, 983–1015.
- Easton, P., Taylor, G., Shroff, P., and Sougiannis, T., 2002. Using forecasts of earnings to simultaneously estimate growth and the rate of return on equity investment. *Journal of accounting research*, 40, 659–676.
- Fama, E. and French, K., 1992. The cross-section of expected stock returns. *Journal of finance*, 42 (2), 427–465.
- Fama, E. and French, K., 1997. Industry costs of equity. *Journal of financial economics*, 43, 153–193.
- Fama, E. and French, K., 2002. The equity premium. *Journal of finance*, 58, 609–646.

- Fisher, D., 2007. The magic metric. *Forbes*, 180 (12), 116–118.
- Foerster, S. and Karolyi, G., 1999. The effects of market segmentation and investor recognition on asset prices. *Journal of finance*, 54, 981–1014.
- Gebhardt, W., Lee, C., and Swaminathan, B., 2001. Toward an implied cost of capital. *Journal of accounting research*, 39, 135–176.
- Gode, D. and Mohanram, P., 2003. Interring the cost of capital using the Ohlson–Juettner model. *Review of accounting studies*, 8, 399–431.
- Gordon, R. and Gordon, M., 1997. The finite horizon expected return model. *Financial analysts journal*, 53, 52–61.
- Guerrera, F., 2006. Why Korea trades at a discount. *Financial times*.
- Habib, A., 2006. Information risk and the cost of capital: review of the empirical literature. *Journal of accounting literature*, 25, 127–168.
- Hail, L., 2002. The impact of voluntary corporate disclosures on the ex-ante cost of capital for Swiss firms. *The European accounting review*, 11 (4), 741–773.
- Hail, L. and Leuz, C., 2006. International differences in the cost of equity capital: do legal institutions and securities regulation matter? *Journal of accounting research*, 44 (3), 485–531.
- Harris, Robert S., Marston, Felicia C., Mishra, Dev R., and O'Brien, Thomas J., 2003. Ex ante cost of equity estimates of S&P 500 firms: the choice between global and domestic CAPM. *Financial management*, 32, 51–66.
- He, Z. and Kryzanowski, L., 2007. Cost of equity for Canadian and US sectors. *The North American journal of economics and finance*, 18, 215–229.
- Henry, P., 2000. Stock market liberalization, economic reform, and emerging market equity prices. *Journal of finance*, 55 (2), 529–564.
- Kaoufmann, D., Kraay, A., and Mastruzzi, M., 2008. Governance matters VII: Aggregate and Individual Governance Indicators, 1996–2008 (29 June, 2009). World Bank Policy Research Working Paper No. 4978. Available at SSRN: <http://ssrn.com/abstract=1424591>
- Khan, M. and Watts, R., 2008. *Estimation and empirical properties of a firm-year measure of conservatism*. Working paper. Massachusetts Institute of Technology.
- Kim, H., 2008. Korea's national brand campaign at crossroads, *The Munhwa Ilbo*, 1 October, p. 38.
- Koedijk, K.K. and van Dijk, M.A., 2004. Global risk factors and the cost of capital. *Financial analysts journal*, 60, 32–38.
- Kumar, S. and Hyodo, K., 2001. Price-earnings ratios in Japan: recent evidence and further results. *Journal of international financial management and accounting*, 12 (1), 24–49.
- Lambert, R., Leuz, C., and Verrecchia, R., 2007. Accounting information, disclosure, and the cost of capital. *Journal of accounting research*, 45 (2), 385–420.
- Lee, C. and Ng, D., 2006. *Corruption and international valuation: Does virtue pay?* Working paper. Cornell University.
- Lintner, J., 1965. Security prices, risk, and maximal gains from diversification. *Journal of finance*, 20, 587–615.
- Miller, D., 1999. The market reaction to international cross-listings: evidence from depository receipts. *Journal of financial economics*, 51 (1), 103–123.
- Moon, G., 2008. IFRS to benefit local firms. *JoongAng Daily*, 17 December, p. 4.
- Mossin, J., 1966. Equilibrium in a capital asset market. *Econometrica*, 34, 768–783.
- Nissim, D. and Penman, S., 2003. The association between changes in interest rates, earnings, and equity values. *Contemporary accounting research*, 20 (4), 775–804.
- O'Hanlon, J. and Steele, A., 2000. Estimating the equity risk premium using accounting fundamentals. *Journal of business, finance and accounting*, 27, 1051–1083.
- Ohlson, J.A., 1995. Earnings, book values, and dividends in equity valuation. *Contemporary accounting review*, 11 (2), 661–687.
- Ohlson, J. and Juettner-Nauroth, B., 2005. Expected EPS and EPS growth as determinants of value. *Review of accounting studies*, 10, 349–365.
- Pantazis, C., Park, J., and Sutton, N., June 2008. Corruption and valuation of multinational corporation. *Journal of empirical finance*, 15 (3), 387–417.
- Park, S., 2000. What does the P–E ratio mean? *Journal of investing*, 9, 27–35 (Fall).
- Poshakwale, S. and Curtis, J., 2005. Disclosure level and cost of equity capital: evidence from the banking industry. *Managerial and decision economics*, 26, 431–444.

- Pyo, J. and Jung, H., 2007. 'Korea Discount' may be fading, helping stocks. *Joongang Daily*, 14 February, p. 3.
- Ramstad, E., 2007a. End to 'Korea Discount'? *The Wall Street Journal*, 16 March, p. 17.
- Ramstad, E., 2007b. What a new market tag could mean for Korea. *The Wall Street Journal Asia*, 19 September, p. 17.
- Richardson, S., Teoh, S., and Wysocki, P., 2000. *Tracking analysts' Forecasts over the annual earnings horizon: optimism, pessimism, and inside trading*. Working paper. University of Michigan and The Ohio State University.
- Sabal, J., 2004. The discount rate in emerging markets: a guide. *Journal of applied corporate finance*, 16, 155–166.
- Salmon, A., 2007. Cutting the Korean discount. *Forbes Asia*, 3 (9), 34–38.
- Seggerman, H., 2007. Korea faces uphill battle to become financial hub. *The Korea Times*, 13 June.
- Shahed, I., Richard, B., and Colin, C., 2008. The use of valuation models by UK investment analysts. *European accounting review*, 17 (3), 503–535.
- Sharpe, W., 1964. Capital asset prices: a theory of market equilibrium under conditions of risk. *Journal of finance*, 19, 425–442.
- Suh, J. and Sim, S., 2007. Korea discount: diagnosis and remedy. *Asia-Pacific journal of financial studies*, 36 (4), 621–655.